

CSE 4018: Structured Programming Lab

Lab 6

Objectives

- Introduction to switch-case
- Loops with break and continue statement
- Introduction to nested loops

Tasks

1. Write a program that reads two numbers from users and performs an operation based on the user's choice (given as a character).
 - Add the numbers if the user provides *a* or *A*
 - Subtract the numbers if the user provides *s* or *S*
 - Multiply the numbers if the user provides *m* or *M*
 - Divide the numbers if the user provides *d* or *D*
 - Print the reverse of the integer part of the first number if the user provides *r* or *R*
[**Example: Reverse of number 123 is 321, Reverse of 57.32 is 75**]
 - Print "Invalid choice" for any other character
 - Print error message saying "Invalid input" if the selected operation can not be performed on provided numbers.

[**NB:** You MUST use switch-case to select the operation based on the user's input.]

2. Modify the task1 so that after every operation, the program asks the user if he wants to continue or not. If the user wants to continue, then the program will ask for two new numbers again and continue with the rest of the operations.

[**Optional task:** In case of invalid input, the program will keep asking to provide the input until the user provides a valid input.]

3. Write a program to implement a simple guessing game. The description and scoring method is provided below:

Rules:

- The program predicts a random number between 1 to 1024 and the user has to GUESS the number.
- The user gets an unlimited number of attempts to guess the number.
- If the user's guess is incorrect, the program informs the user, whether the given number is higher or lower than the predicted number.
- The game ends when the user successfully guesses the predicted number.

Scoring:

- The score of the game is determined by the number of attempts the user needs to guess the number.
- After each game, the program asks the user whether he wants to continue the game or wants to quit.
- Once the user quits the game, the program will print the **average** score for all the games the user has played so far.

An executable file (**Guessing_Game.exe**) is provided as an attachment to the Google Classroom task.

Resource: The code for generating a random number is provided below:

```
#include <time.h>
#include <stdlib.h>

srand(time(NULL));
int r = rand();
```